

FUNCTIONAL AND NON- FUNCTIONAL REQUIREMENTS

NAME: BHADRA R S

SRN: PES1UG24CS111

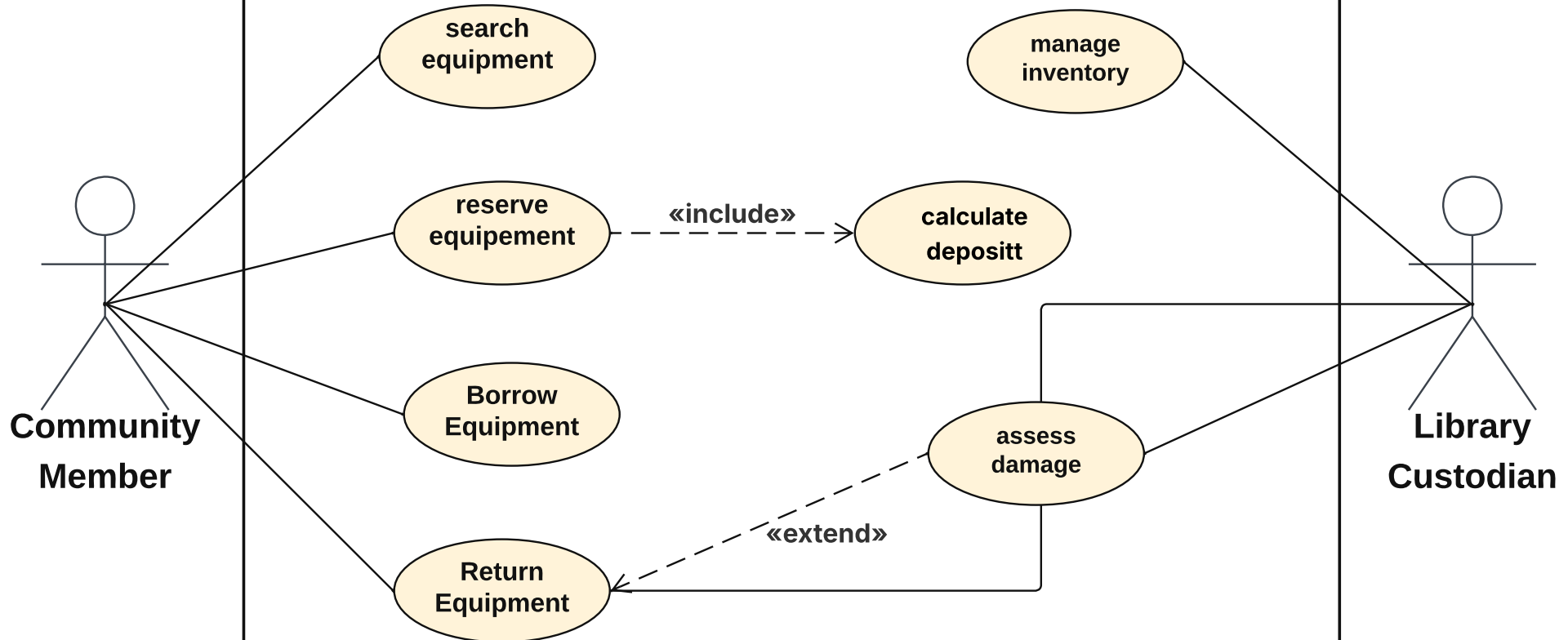
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	
FR-001	Functional	The system shall calculate required security deposits based on tool category, enforce loan duration limits, and calculate late return fees.	High	Pass: For a selected tool, the system calculates the correct deposit, enforces the maximum loan duration, and calculates a late fee when the equipment is returned late.	Ensures deposits, borrowing periods, and late fees are handled correctly.	
FR-002	Functional	The system shall allow community members to search for equipment by name or category and view its availability.	High	Pass: When a member searches for an equipment name or category, the system displays matching equipment and its current availability.	Allows members to find equipment and determine whether it is available.	
FR-003	Functional	The system shall allow community members to reserve	High	Pass: When a member selects available equipment	Allows members to reserve equipment before	

		available equipment for a selected borrowing period.		and a valid borrowing period, the system creates and displays a reservation confirmation.	borrowing it.	
FR-004	Functional	The system shall record the checkout and return of equipment and update its availability status.	High	Pass: After equipment is checked out, its status changes to unavailable, and after it is returned, its status changes to available.	Keeps equipment availability accurate.	
FR-005	Functional	The system shall allow the library custodian to record damage found during equipment return inspection.	Medium	Pass: When damage is identified during a return inspection, the custodian can record the damage details in the system.	Provides a record of equipment damage after use.	
NFR-001	Nonfunctional	The tool inventory search must display real-time physical availability across neighborhood depot lockers.	High	Pass: When equipment is checked out or returned, its availability shown in inventory search reflects the updated physical availability.	Ensures members and custodians see accurate equipment availability.	
NFR-002	Nonfunctional	The system shall display equipment search results within 2 seconds under normal	Medium	Pass: At least 95% of equipment searches return results within 2	Ensures that members can quickly find equipment.	

		operating conditions.		seconds during normal operating conditions.		
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Community Tool & Equipment Library System

BHADRA R S PES1UG24CS111



Use-Case Flow Specification

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Use Case: Reserve Equipment

1. Preconditions

1. The community member is registered and logged into the system.
2. The requested equipment exists in the library inventory.
3. The equipment is currently available.

2. Postconditions

1. The equipment is reserved for the community member.
2. The reservation is recorded in the system.
3. The required security deposit is calculated and recorded.

3. Main Success Scenario

1. The community member searches for the required equipment.
2. The system displays matching equipment and its availability.
3. The community member selects an available equipment item.
4. The community member specifies the required borrowing period.
5. The system checks that the selected borrowing period is within the allowed loan duration.
6. The system calculates the required security deposit.
7. The system displays the reservation and deposit details.
8. The community member confirms the reservation.
9. The system records the reservation.
10. The system displays a reservation confirmation.

4. Alternate Flow — Equipment Unavailable

- 3a. The selected equipment is no longer available.
 - 3a1. The system informs the community member that the equipment is unavailable.
 - 3a2. The system does not create the reservation.
 - 3a3. The community member may search for and select another available equipment item.

